

MAYANK BANSAL

Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Second-year Computer Science student at LPU specializing in applied machine learning and real-world system development. Built and deployed multiple end-to-end ML systems across NLP, computer vision, and vector search, including a RAG-based question answering system and a visual search engine. Experienced in designing scalable pipelines, working with embeddings, and integrating ML models into production-ready APIs. Published researcher with a 9.33 CGPA and 580+ DSA problems solved.

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, CNNs, Neural Networks, Feature Engineering, Model Evaluation, EDA

Vector Search & RAG: FAISS, Sentence Transformers, Retrieval Augmented Generation

Backend & Deployment: Flask, FastAPI, REST APIs, Docker

Data & Languages: Python, NumPy, Pandas, Matplotlib, Seaborn, SQL

PROJECTS

AI Complaint Triage & Routing System - Python, Flask, Scikit-learn, TF-IDF, SQLite | [GitHub](#)

Jan 2026 – Feb 2026

- Automated complaint routing across 8 departments with urgency prediction using a hybrid credibility scoring pipeline combining rule-based logic and ML confidence scores
- Designed and published an 800-sample multi-class complaint dataset on Kaggle; built a modular Flask REST API with explainable decision flow and a foundation for continuous retraining

Visual Search Engine - PyTorch, FAISS, FastAPI, JavaScript | [GitHub](#)

Feb 2026 – Mar 2026

- Built an image-based retrieval system using a pretrained ResNet50 feature extractor generating 2048-dim embeddings and FAISS nearest-neighbour search over a fashion dataset, delivering real-time visual similarity results
- Engineered a FastAPI backend integrating image preprocessing, embedding generation, and vector search with a responsive upload-and-preview frontend — full end-to-end deployment

PDF Question Answering System (RAG) - FAISS, Sentence Transformers, Flask, Gemini API | [GitHub](#)

Dec 2025 – Jan 2026

- Implemented a RAG pipeline with document chunking, FAISS vector similarity search, and Sentence Transformer embeddings to ground LLM responses in retrieved PDF context
- Engineered prompt-level guardrails to reduce hallucinations and enforce explicit out-of-scope responses; delivered transparent answers with source attribution via Flask + HTML/CSS/JS frontend

Mood2Mail — Real-Time Email Tone Analyzer - TF-IDF, Naive Bayes, Flask, JavaScript | [GitHub](#)

Apr 2025

- Achieved 96.84% accuracy (weighted F1: 0.97) across 5 tone categories on a manually annotated 475-sample email dataset published on Kaggle; research paper published on Zenodo with DOI
- Deployed real-time tone feedback via a responsive Flask + JavaScript interface, enabling proactive tone correction before sending

ACHIEVEMENTS & CERTIFICATIONS

- Published a technical book** — "Machine Learning Projects: A Practical Guide for B.Tech CSE Students," on Amazon Kindle and Paperback, covering 6 end-to-end ML projects across NLP, CNNs, Transfer Learning, and Audio ML (<https://www.amazon.com/dp/B0GX34T5HF>)
- Published research paper** — "Mood2Mail: A Lightweight Real-Time Email Tone Analyzer Using Classical ML" on Zenodo with DOI
- Competitive programming** — 580+ problems on GeeksForGeeks (101-day streak, rank 159 / 39,000+ participants) + 100+ on LeetCode
- Certifications** — Complete Data Science & ML, GeeksForGeeks (Jun–Nov 2025) - AI Tools Skill Up, GeeksForGeeks (Jun–Aug 2025)

EDUCATION

Lovely Professional University — B.Tech Computer Science Engineering

Aug 2024 – May 2028

CGPA: 9.33/10 - Phagwara, Punjab, India